

Phase 4 - Backend Deployment on Render (Detailed Training Manual)

Project: Bitto MERN Website

Objective: Deploy the Node.js backend on Render, connect it with MongoDB Atlas, troubleshoot deployment issues, and verify the production API.

Step 1 - Verify Backend Locally

Run: `npm run dev`

Verify MongoDB connected and API running before deployment.

Step 2 - Verify Health API

Open `http://localhost:5000/api/health` and verify JSON response.

Step 3 - Login to Render

Sign in using GitHub.

Step 4 - Create Web Service

New → Web Service.

Step 5 - Connect Repository

Select `bitu_website` repository.

Step 6 - Configure Service

Root Directory: `backend`

Build Command: `npm install`

Start Command: `npm start`

Branch: `master`

Step 7 - Add Environment Variables

Add `PORT`, `MONGO_URI`, `JWT_SECRET`, `ADMIN_NAME`, `ADMIN_EMAIL`, `ADMIN_PASSWORD`, `CLIENT_URL`.

Step 8 - Deploy

Click Deploy Web Service.

Step 9 - Fix MONGO_URI Error

Added missing environment variables and redeployed.

Step 10 - Fix Atlas Connection

Added `0.0.0.0/0` in MongoDB Atlas Network Access.

Step 11 - Successful Logs

MongoDB connected

API running on port 5000

Step 12 - Verify Production

Open <https://<render-url>/api/health> and verify response.

Errors Encountered

1. MONGO_URI undefined

Root Cause: Environment variables were missing.

Solution: Added variables and redeployed.

2. Could not connect to MongoDB Atlas

Root Cause: Render IP wasn't allowed.

Solution: Added 0.0.0.0/0 to Atlas Network Access.

3. Route not found

Root Cause: Opened '/' instead of '/api/health'.

Solution: Used the correct endpoint.

Final Architecture

Developer

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Git Push

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GitHub

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Render Backend

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MongoDB Atlas

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Client (HTTPS)

What I Learned From This Phase

Learned how to deploy a Node.js backend on Render.

Learned how Render connects with a GitHub repository.

Learned the difference between Build Command and Start Command.

Learned why the Root Directory is important during deployment.

Learned how package.json controls the deployment process.

Learned how to use Environment Variables in production.

Learned why sensitive credentials should not be stored in GitHub.

Learned how to connect Render with MongoDB Atlas.

Learned why MongoDB Atlas Network Access (IP Whitelisting) is required.

Learned how to troubleshoot deployment errors using Render logs.

Learned how to verify the backend using the Health API.

Learned the complete deployment workflow: GitHub → Render → MongoDB Atlas.